

SW Engineering CSC648-848 Fall 2025

GatorsMatch Tutors

Team 01

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1.) Executive Summary

GatorMatch Tutors is an SFSU-only web application that makes it fast, safe, and credible for students to find peer tutors for the exact courses they're taking. Instead of bouncing across generic marketplaces and informal channels, any @sfsu.edu student can search by course code (e.g., CSC 340), see meeting options at a glance (Zoom or on-campus), review concise tutor profiles, and send a single in-site message to start contact—no email chase or chat threads. Every tutor profile clearly surfaces trust signals that matter to students and faculty: a 5-star rating generated by verified students who have used that tutor's services, the total hours tutored to date, and an eligibility badge confirming the tutor has completed the course with a minimum grade of 90% and/or holds a professor's recommendation. Tutors can create and manage listings with clear status (Draft, Pending, Approved), and every listing is reviewed by an administrator who can approve, reject with a reason, or later unpublish to keep the catalog clean and trustworthy.

What makes GatorMatch distinctive is its SFSU fit. Discovery is course-centric and campus-aware: listings are tagged with official course codes and campus locations such as the J. Paul Leonard Library, and availability is expressed in semester-friendly terms students actually use ("post-lab hours," "midterm week evenings"). Filters are data-driven so options with zero matches never waste a student's time. The result is a moderated, auditable directory that shortens time-to-help for students, channels relevant inquiries to tutors, and provides the university a safe, focused service without payments or external messaging.

2.) List of All Data Items & Entities

- Key Users

- Unregistered Student

Definition	Any user who visits the platform without logging in
Permissions	Can browse and search tutor listings, but cannot send messages, create or edit listings, or access dashboards
Usage	Enables casual browsing before sign-up, encouraging new users to register once they find relevant tutors

- Registered Student

Definition	A verified SFSU student logged into the system to find tutoring help
Permissions	Can browse/search tutor listings and send a one-time message to tutors
Usage	Primary consumer of tutoring services; uses the platform to find and contact peer tutors

- Registered Tutor

Definition	A verified SFSU student providing tutoring services through the platform
Permissions	Can create and update tutor listings; submit listings for admin approval
Usage	Primary provider of tutoring services; maintains visibility and availability through listings

- Admin

Definition	A user responsible for managing and moderating the platform
Permissions	Can view all tutor listings, approve or reject posts (with reasons), unpublish or delete content, and moderate users
Usage	Ensures platform quality, verifies listings, and maintains a safe environment for all users

- Data Items

- User Profile

Definition	Stores verified SFSU user information and determines the user's role
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	(Student, Tutor, or Admin)
Sub Items	• Full Name • SFSU Email (@sfsu.edu) • Role (Student, Tutor, Admin) • Profile Picture (optional) • Registration Date • Account Status (Active / Blocked)
Input	Created during registration and updated by the user
Usage	Used for authentication, access control, and displaying role-based dashboards

○ Tutor Profile

Definition	A profile-like listing created by a tutor that advertises their services
Sub Items	• Tutor Name • Subject Expertise (e.g., CSC 648, MATH 226) • Availability (day and time) • Campus Location Tag (Library, Zoom, etc.) • Additional Notes / Description • Status (Pending, Approved, Rejected) • Admin Review Reason (optional)
Input	Created and submitted by Tutor; requires Admin approval before being public
Usage	Displayed in searchable listings for Students; visibility controlled by approval status

○ Message

Definition	A one-round, in-site message sent from a Student to a Tutor regarding a specific listing
Sub Items	• Message ID • Sender (Student) • Receiver (Tutor) • Linked Listing ID • Message Content • Timestamp • Status (Sent, Read)
Input	Created automatically when a Student sends a one-round message to a Tutor
Usage	Appears in the Student's "Sent" box and Tutor's "Inbox"; no replies or external email allowed

○ Admin Review Log

Definition	A record of all Admin actions taken on Tutor Posts or Messages for accountability and moderation
Sub Items	• Admin ID • Action Type (Approve, Reject, Unpublish, Delete) • Target Post / Message ID • Timestamp • Reason / Notes
Input	Automatically generated when Admin performs an action
Usage	Enables transparent moderation tracking and supports audit or rollback if needed

○ Search Tutors

Definition	Store search preferences from students to find matching tutors.
Sub Items	• Course Name • Keyword • Subject • Tutor's Profile • Tutor's Rating
Input	From the student's search form.
Usage	Filters Tutor data (profile, availability, reviews) to show ranked results.

○ Become a Tutor

Definition	Record information tutors provide when applying to offer sessions.
Sub Items	• Account / Identity • Tutor Profile • Course You Can Tutor • Availability • Verification Documents • Policies
Input	Submitted in "Become a Tutor" form.
Usage	Enables tutor profile publication and matching with student requests.

3.) Functional Requirements - Prioritized

Priority 1:

- **FR-1.1 SFSU Registration:** Users must register with @sfsu.edu (suffix check only).
 - Registered Users
 - Unregistered Users
 - Admin
- **FR-1.2 SFSU Sign-in:** Users must sign in with @sfsu.edu (suffix check only).
 - Registered Users
 - Unregistered Users
 - Admin
- **FR-5.1 Browse & Search:** Search for and browse different tutors by using the search function; options are data-driven and disabled/hidden when there are 0 matches.
 - Registered Users
 - Unregistered Users
- **FR-5.2 Filter:** Filter by Term, Session, Subject, Course Number, Location; options are data-driven and disabled/hidden when there are 0 matches.
 - Registered Users
 - Unregistered Users
- **FR-3 Create/Update Listing:** Tutor can add/edit headline, bio, subjects, course prefixes, availability, location tags, optional display-only rate.
 - Registered Users
- **FR-2 Tutor Role Enablement:** Students can enable a Tutor role to create listings.
 - Registered Users
- **FR-6 Listing Detail:** Detail page shows bio, subjects, course prefixes, availability, location tags, and Contact Tutor.
 - Registered Users
 - Unregistered Users
- **FR-7 One-Round In-Site Messaging:** Students can send one message to a tutor; no replies; no email clients.
 - Registered Users
- **FR-9 SFSU-Specific Discovery:** Course-centric filters, campus location tags, and semester-aware availability text.
 - Registered Users
 - Unregistered Users
- **FR-8 Role-Based Dashboards:** Student (Sent, profile); Tutor (Listings, Inbox, profile); Admin (Approval Queue, Moderation).
 - Registered Users
 - Admin
- **FR-4.1 Admin Approval:** Admin can approve/reject(with reason) certain listings and content.

Priority 2:

- **FR-4.2 Admin Moderation:** Admin can unpublish, delete listings; moderate users/content.
 - Admin

Priority 3:

- **FR-10 Analytics & Banner:** Google Analytics enabled; exact required banner text on every page.
 - Admin

4.) UI Storyboards

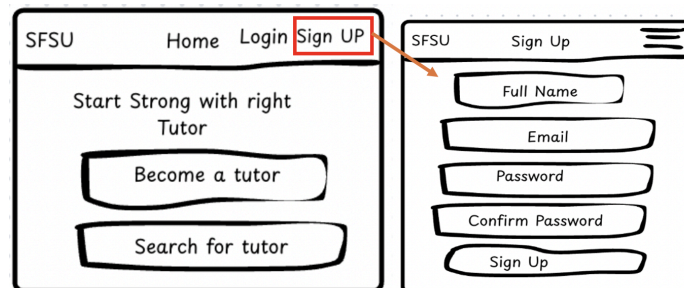
Use Case 1: User Registration

Description:

A new SFSU student or tutor creates an account to access the tutoring platform.

Storyboard (Mockups):

1. **Home Page** – The “Sign up” button is visible on the navbar.
User clicks “Sign up” to create a new account.
2. **Sign up Page** – form includes:
 - Full Name
 - SFSU Email
 - Password / Confirm Password
3. *User submits form → system validates input → sign up success.* **Success Page** – “Sign up successful! don’t need to login”



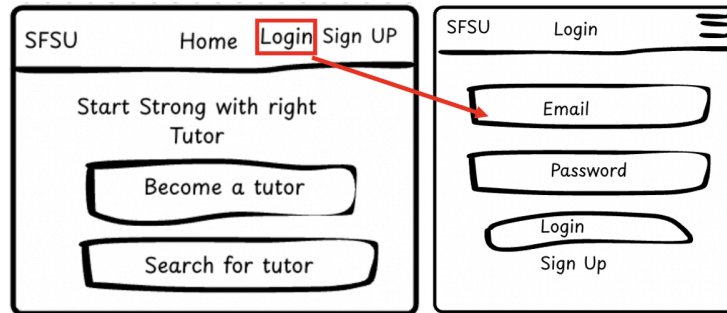
Use Case 2: User Login

Description:

Registered user logs in to access the tutoring dashboard.

Storyboard (Mockups):

1. **Login Page** – SFSU email + password input fields.
User clicks “Login.”
2. **Dashboard Page** – personalized homepage showing upcoming sessions or available tutors.



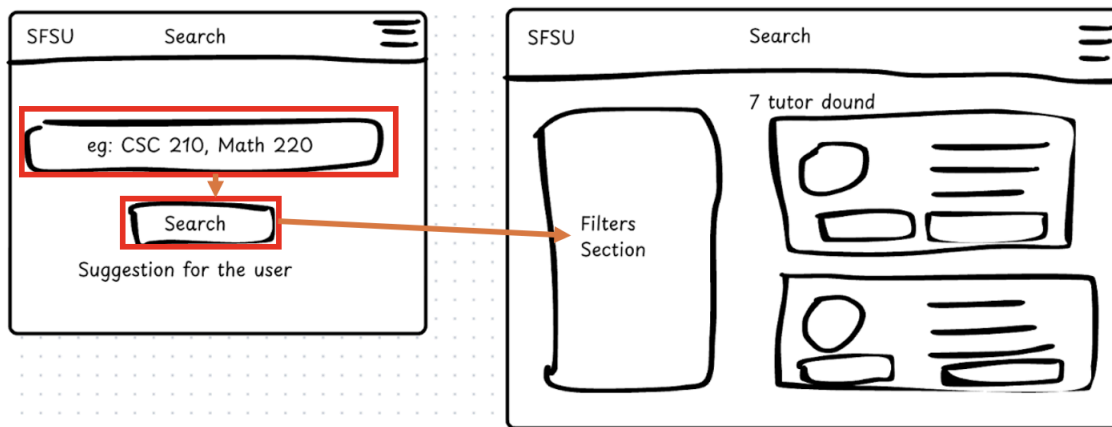
Use Case 3: Browse/Search for Tutors

Description:

Students can view available tutors and filter by subject, availability, or rating.

Storyboard (Mockups):

1. **Dashboard/Home Page** – “Search Tutors” button.
2. **Tutor Listing Page** – list of tutors with name, subject, availability, and “Book” button.
Optional: Apply filters according to the user's needs.



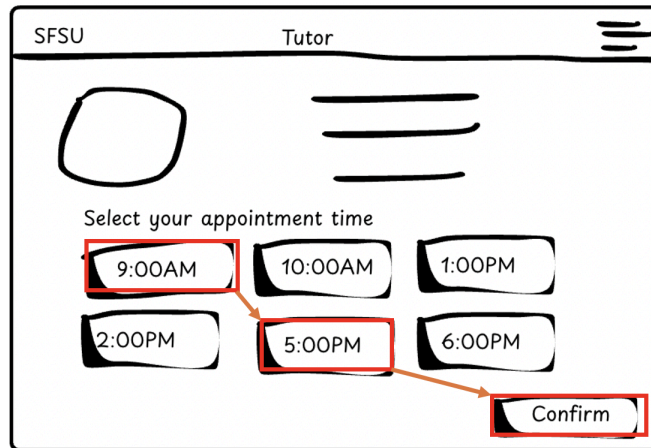
Use Case 4: Book a Tutoring Session

Description:

Students book a time slot with a tutor.

Storyboard (Mockups):

1. **Tutor Profile Page** – tutor’s info and schedule displayed.
User selects a time slot and clicks “Book.”
2. **Booking Confirmation Page** – shows selected date/time and confirmation button.
3. **Success Page** – “Your session has been booked successfully.”



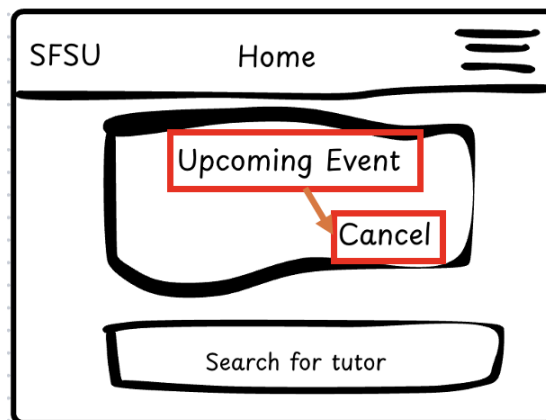
Use Case 5: View Upcoming Sessions (Dashboard)

Description:

Users view and manage their scheduled sessions.

Storyboard (Mockups):

1. **Dashboard Page** – table/list of sessions with columns:
 - Tutor Name
 - Date & Time
 - Subject
 - “Cancel” button (if allowed)
2. *The user can navigate to tutor details or cancel a session.*



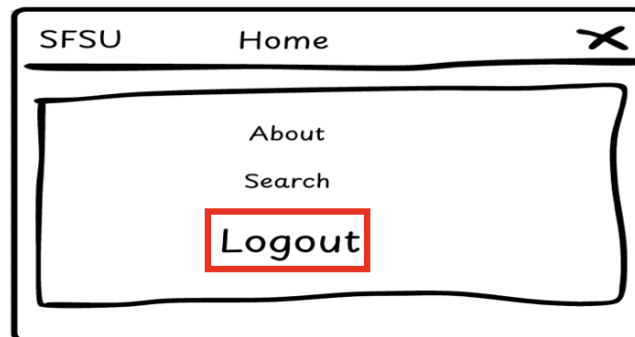
Use Case 6: Logout

Description:

User logs out of the system securely.

Storyboard (Mockups):

1. **Navbar/Menu** – “Logout” option available.
User clicks “Logout.”
2. **Redirect Page** – “You have been logged out” message with link to login page.



5.) High level Architecture, Database Organization summary only

The main tables that we will have in our database will be: User, TutorApplication, and Tutors (searchable).

For the user table, the items / columns we will have are: user id, name, email, password, user role, and registration date.

For the TutorApplication table, the columns we will store are application id, user id, headline, bio, meeting options, courses, availability, documents, application status, created at

For the Tutors table, we will have the following columns: id, user id, application id, slug, headline, bio, meeting options, courses. We will also have some columns about the tutor and their experience: hours tutored, rating, and how many ratings they have received.

Our media (pictures / videos) will be kept in the file system, specifically in the static/images folder.

For searching / filtering the database, we are using LIKE substring matching. It will return the closest match across slug/headline/bio/courses/meeting options. You can also look for tutors within a certain subject prefix or the exact course / location.

6.) Key Risks

Technical Risk:

Our main risk is ensuring the website is fully mobile responsive, since most of the current development focused on desktop-first UX design. Adapting layouts and components for smaller screens may require additional redesign and testing effort.

Resolution:

Adopt a hamburger menu below the md breakpoint to collapse non-critical links, while keeping 1–2 primary actions visible.

Schedule Risk:

Several members of ours are juggling other courses with overlapping deadlines, work or personal matters that could delay completion of milestones or managing work more efficiently.

Resolution:

Use a reliable task manager and check-ins on Fridays to surface any issues early as well as ensure fair production from every member

Skills Risk:

Some team members are still learning or are new to using Flask-MySQL integration and conventions, which may slow our backend process and deployment setup.

Resolution:

Backend lead and support will communicate frequently to align on process as well as reach out to other teammates for input or shared troubleshooting when needed. They'll also use online tutorials and some use of GenAI tools for self learning to strengthen our backend implementation.

Technical/Teamwork Risk:

While milestone 2 involves limited parallel work/development, later milestones will require multiple of our members to be contributing code simultaneously, which could lead to some merge conflicts or even working on the same things others might have already done or are doing at the same time.

Resolution:

Continue using clear branch naming and pull request reviews. A lot will fall on our github manager but active communication between the frontend and backend leads is expected to help prevent conflicts and redundant work. As well as to keep field names and folder structures consistent.

Legal/Content Risk:

Tutor photos or media used in mockups and website could unintentionally violate copyright or student privacy if sourced externally or without consent.

Resolution:

Use royalty free placeholder images from reliable sources, SFSU generated visuals, or our own generated visuals. Frontend team will be responsible for verifying that all visual assets used in the site comply with these norms before submission.

UX / Usability Risk:

If our interface or flow is confusing, users might get frustrated or not want to continue using the site especially when it comes to trying to find a tutor or complete basic options.

Resolution:

Have team members test the mockups and early builds to spot any confusing layouts or even any areas to improve. That way small design adjustments can be given to the front end team to fix based on that feedback. Ensure it's a smooth experience for users by acting like one yourself.

7.) Project Management

ClickUp will function as a unified dashboard for each team member to view and track the status of all assigned tasks. Every task has an owner, due date, priority, a checklist lifted from the M2 PDF, and grouped by the M2 sections (Data Items, Functional Reqs, UI Storyboards, Architecture/DB) with applicable dependencies so the critical path is clear. The status flow of each task will follow 'To Do → In Progress → In Review → Done'. Team lead will review each task prior to moving the status to 'Done'. Primary communication will be conducted via discord, with regular check-ins prior to the weekly team meeting on Friday.

The front end owns: the page templates (Jinja/HTML), the visual design (CSS), and small, focused JavaScript for UI details (the hamburger menu, chips, counters). The back end owns: Flask routes and APIs, database models and queries (SQLAlchemy), authentication and sessions, validation, file uploads, search/filter/pagination, and security. Common interfaces will be agreed upon (each page or API we agree on the URL, inputs, and the exact field names returned to templates/JSON (matching the data names defined in Section 2).

8.) Use of GenAI

For Milestone 2, our team's use of Generative AI (GenAI) tools was limited to light research and clarification support, rather than direct content generation. Most written content, including requirements, UI storyboards, and architecture descriptions, was drafted collaboratively through group discussion and manual editing. However, we recognize the potential of GenAI tools (such as ChatGPT, GitHub Copilot, or others) in supporting different aspects of the software engineering lifecycle, and we plan to integrate them more actively in upcoming milestones.

Specifically, we anticipate using GenAI in the following ways:

- Code Assistance & Debugging: In order to write and fix small portions of our code more quickly, we intend to use GenAI tools.
- Documentation & Comments: For our project, GenAI can assist us in creating concise documentation and clear comments.
- UI/UX Feedback and Copy Suggestions: To enhance the user experience, we might employ GenAI to recommend better wording or layouts.
- Brainstorming & Risk Management: GenAI can help us generate innovative ideas and identify potential project risks early.

9.) Team Lead Checklist

Checklist Item	Status	Comments
So far, all team members are fully engaged and attending team sessions when required	OK	Consistently engaged via discord &

		attending classes.
Team reviewed suggested resources before drafting Milestone 2	OK	
Team lead checked Milestone 2 document for quality, completeness, formatting and compliance with instructions before the submission	OK	
Team lead ensured that all team members read the final Milestone 2 document and agree/understand it before submission	OK	
Team shared and discussed experience with GenAI tools among themselves	OK	